EXERCISE 1

```python
# Read file
with open("logins.txt", "r") as file:
    names = file.read().splitlines()

# 1. Print all names
print("All Names:", names)

# 2. Total records
print("Total Logins:", len(names))

# 3. Frequency count
count = {}
for name in names:
    if name in count:
        count[name] += 1
    else:
        count[name] = 1
```

```
"C:\Users\azhagu murugan S\PycharmProjects\PythonProject\.venv\Scripts\python.exe" "C:\Users\azhagu murugan S\PycharmProjects\PythonProject\apr18ex1.py"
All Names: ['Rahul', 'Sneha', 'Rahul', 'Arjun', 'Sneha', 'Rahul', 'Karan']
Total Logins: 7
Login Count: {'Rahul': 3, 'Sneha': 2, 'Arjun': 1, 'Karan': 1}
Most Active User: Rahul
Unique Users: {'Sneha', 'Arjun', 'Karan', 'Rahul'}

Process finished with exit code 0
```

EXERCISE 2

```python
# Sum
print("Sum:", sum(numbers))

# Max
print("Max:", max(numbers))

# Min
print("Min:", min(numbers))

# Count > 50
count = 0
for num in numbers:
    if num > 50:
        count += 1

print("Greater than 50:", count)
```

```
"C:\Users\azhagu murugan S\PycharmProjects\PythonProject\.venv\Scripts\python.exe" "C:\Users\azhagu murugan S\PycharmProjects\PythonProject\apr18ex2.py"
Numbers: [45, 67, 23, 89, 12, 90, 56, 34, 78, 21]
Sum: 515
Max: 90
Min: 12
Greater than 50: 5

Process finished with exit code 0
```

EXERCISE 3

```python
top = max(students, key=lambda x: x["marks"])
print("Topper:", top["name"])

# 4. Average
total = sum(s["marks"] for s in students)
print("Average:", total / len(students))

# 5. Course count
course_count = {}
for s in students:
    course = s["course"]
    if course in course_count:
        course_count[course] += 1
    else:
        course_count[course] = 1

print(course_count)
```

```
Python Students:
Rahul
Arjun
Karan
Topper: Sneha
Average: 82.6
{'Python': 3, 'Data Science': 1, 'AI': 1}

Process finished with exit code 0
```

EXERCISE 4

```python
import json

with open("orders.json", "r") as file:
    data = json.load(file)

orders = data["orders"]

# 1. Print all orders
print(orders)

# 2. Total revenue
total = sum(o["amount"] for o in orders)
print("Revenue:", total)

# 3. Spending per customer
spending = {}
for o in orders:
    c = o["customer"]
```

```
"C:\Users\azhagu murugan S\PycharmProjects\PythonProject\.venv\Scripts\python.exe" "C:\Users\azhagu murugan S\PycharmProjects\PythonProject\apr18ex4.py"
[{'order_id': 1, 'customer': 'Rahul', 'amount': 2500}, {'order_id': 2, 'customer': 'Sneha', 'amount': 1800}, {'order_id': 3, 'customer': 'Rahul', 'amount': 3200}, {'order_id':
Revenue: 11100
{'Rahul': 5700, 'Sneha': 3900, 'Arjun': 1500}
Top Customer: Rahul
{'Rahul': 2, 'Sneha': 2, 'Arjun': 1}

Process finished with exit code 0
```

EXERCISE 5

```python
import csv

with open("employees.csv", "r") as file:
    reader = csv.DictReader(file)
    employees = list(reader)

# 1. Names
for emp in employees:
    print(emp["name"])

# 2. IT department
print("IT Employees:")
for emp in employees:
    if emp["department"] == "IT":
        print(emp["name"])

# 3. Average salary
total = sum(int(emp["salary"]) for emp in employees)
```

```
IT Employees:
Rahul
Arjun
Karan
Average: 71400.0
Highest Paid: Priya
{'IT': 3, 'HR': 1, 'Finance': 1}

Process finished with exit code 0
```

EXERCISE 6

```python
import csv

with open("sales.csv", "r") as file:
    reader = csv.DictReader(file)
    sales = list(reader)

# 1. Total revenue
total = 0
for s in sales:
    total += int(s["quantity"]) * int(s["price"])
print("Total Revenue:", total)

# 2. Quantity per product
qty = {}
for s in sales:
    p = s["product"]
    q = int(s["quantity"])
    if p in qty:
```

```
"C:\Users\azhagu murugan S\PycharmProjects\PythonProject\.venv\Scripts\python.exe" "C:\Users\azhagu murugan S\PycharmProjects\PythonProject\apr18ex6.py"
Total Revenue: 637500
{'Laptop': 8, 'Mouse': 30, 'Keyboard': 15}
Top Product: Mouse
{'Laptop': 600000, 'Mouse': 15000, 'Keyboard': 22500}
Laptop

Process finished with exit code 0
```

BONUS

```python
import csv

with open("sales.csv", "r") as file:
    reader = csv.DictReader(file)
    sales = list(reader)

# 1. Total revenue
total = 0
for s in sales:
    total += int(s["quantity"]) * int(s["price"])
print("Total Revenue:", total)

# 2. Quantity per product
qty = {}
for s in sales:
    p = s["product"]
    q = int(s["quantity"])
    if p in qty:
```

```
"C:\Users\azhagu murugan S\PycharmProjects\PythonProject\.venv\Scripts\python.exe" "C:\Users\azhagu murugan S\PycharmProjects\PythonProject\apr18ex6.py"
Total Revenue: 637500
{'Laptop': 8, 'Mouse': 30, 'Keyboard': 15}
Top Product: Mouse
{'Laptop': 600000, 'Mouse': 15000, 'Keyboard': 22500}
Laptop

Process finished with exit code 0
```